

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER-VII EXAMINATION – WINTER 2025

Subject Code:3174205

Date:01-12-2025

Subject Name:Natural Language Processing

Time:10:30 AM TO 01:00 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) What is Natural Language Processing (NLP)? Why is NLP considered a difficult task?	03
	(b) List out the advantages and disadvantages of NLP.	04
	(c) Explain the different phases of NLP in detail.	07
Q.2	(a) What is morphology in the context of Natural Language Processing?	03
	(b) What is the role of smoothing in language modeling? Provide an example of why it is needed.	04
	(c) Explain the importance of Named Entity Recognition (NER) in NLP. How does it work, and what challenges are faced when implementing the NER system?	07
OR		
	(c) Discuss the differences between unigram, bigram, and trigram models. How do they capture different levels of context in language?	07
Q.3	(a) How does Part of Speech Tagging work in NLP?	03
	(b) Draw and explain the architecture of the Natural Language Generation (NLG) system.	04
	(c) Explain how the skip-gram model works and discuss its advantages in capturing semantic relationships between words.	07
OR		
Q.3	(a) Briefly describe the role of Lexical Semantics in NLP.	03
	(b) What is the Bag of Words model, and how is it used in NLP?	04
	(c) Explain the process of Word Sense Disambiguation and compare the Knowledge-Based and Supervised approaches, providing examples of how each method handles the disambiguation task.	07
Q.4	(a) State different algorithms used for relation extraction.	03
	(b) Briefly explain the role of Named Entity Recognition in Information Extraction.	04
	(c) Explain the different phases of an Information Retrieval based question-answering model.	07

OR

- Q.4** (a) What is Sentiment Mining, and how is it applied in NLP? **03**
(b) Define the following terms: Text Classification and Text Summarization. **04**
(c) Discuss the role of NLP in Information Retrieval (IR). How does Cross-Lingual IR differ from traditional IR, and what are some challenges associated with it? **07**
- Q.5** (a) What is the Rule-Based Machine Translation (RBMT) method? **03**
(b) Explain the encoder-decoder model of machine translation. **04**
(c) Explain with an illustration how parameter learning in Statistical Machine Translation (SMT) is conducted. **07**

OR

- Q.5** (a) What are the advantages of using neural machine translation over the traditional machine translation approach? **03**
(b) Briefly discuss Statistical Machine Translation (SMT). **04**
(c) How Direct Machine translation approach work in NLP? **07**
